ADVANCED ENERGY MATERIALS
www.advenergymat.de

A Graphene-Coated Thermal Conductive Separator to Eliminate the Dendrite-Induced Local Hotspots for Stable Lithium Cycling

Duzhao Han, Xiaowei Wang, Ya-Nan Zhou, Jiyong Zhang, Zhongxin Liu, Zichun Xiao, Jiangqi Zhou, Zhen Wang, Jiangfeng Zheng, Zhanhui Jia, Bingbing Tian, Jingying Xie, Zhaolin Liu,* and Wei Tang**

Practical lithium metal batteries (LMBs) are still far from market readiness, as a result of the severe Li degradation and safety issues caused by Li dendrites. Herein, by studying the thermodynamic behavior of lithium deposition, it is unveiled that the tip area of Li metal has an increasing heat generation rate as a function of the deposition time and overpotential. This triggers the emergence of the accumulated overpotential heat and local temperature "hotspots" due to poor local thermal diffusion, which exacerbates the undesirable irregular Li deposition and dendrite growth. To address this issue, a thermally conductive graphene-coated separator is constructed to eliminate these local hotspots. The graphene layer affords timely diffusion of local heat generated by irregular Li growth and incipient dendrite formation, achieving the stable and uniform lithium deposition to deter further degradation. As a result, the Li metal, suffering a drastic Coulombic efficiency (CE) decay to ≈60% using a conventional separator, can be recovered for continual cycling with a high CE of >95%. Notably, the corresponding Li||$LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ cells present high capacity retention and recovery. This study highlights the thermodynamic factor of Li dendrite-induced local heat and its elimination to preclude Li anode deterioration, which provides insight into Li metal protection strategies for high performance LMBs.

1. Introduction

The demand of high-energy-density applications like electric vehicles and grid energy storage applications has exceeded the capabilities of state-of-the-art batteries, such as lithium-ion batteries (LIBs) based on the commercial graphite anode with a limited theoretical specific capacity of ≈372 mAh g^{-1}.[1] Rechargeable lithium metal batteries (LMBs) are considered to be the key to next generation high-energy-density batteries, since Li metal anode presents a theoretical gravimetric specific capacity up to 3860 mAh g^{-1} and a low reduction potential of −3.04 V versus standard hydrogen electrode (SHE).[2] However, the applicability of LMBs is severely hindered by a series of bottlenecks caused by lithium anode, including dendritic lithium growth, unstable solid-electrolyte interphase (SEI), and almost infinite volume change during cycling. Among these, the inevitable dendrite formation and growth is one of the most serious issues, which aggravates the adverse side-reaction and the formation of "dead Li". Consequently, it leads to low Coulombic efficiency (CE), limited life span, and severe safety risks.[2] To seize the "holy grail" of Li metal anode, the research community significantly focuses on how to maintain the flat surface morphology and dense structure of the metallic Li deposition without apparent dendrites during the electrochemical plating/stripping processes.

D. Han, Y.-N. Zhou, Z. Xiao, J. Zhou, Z. Wang, W. Tang
School of Chemical Engineering and Technology
Xi'an Jiaotong University
Xi'an, Shaanxi 710049, China
E-mail: tangw2018@mail.xjtu.edu.cn

X. Wang, Z. Liu
Institute of Materials Research and Engineering
A*STAR
2 Fusionopolis Way
Innovis 138634, Singapore
E-mail: zl-liu@imre.a-star.edu.sg

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/aenm.202201190.

DOI: 10.1002/aenm.202201190

J. Zhang, Z. Liu, J. Xie
State Key Laboratory of Space Power-Sources Technology
Shanghai Institute of Space Power Sources
Shanghai 200245, China
E-mail: jyxie@mail.sim.ac.cn

J. Zheng
Guangdong Jiana Energy Technology Co., Ltd.
Guangzhou 511449, China

Z. Jia
State Key Laboratory for Mechanical Behavior of Materials
Xi'an Jiaotong University
Xi'an, Shaanxi 710049, China

B. Tian
Shenzhen University
Shenzhen 518000, China

 © 2022 Wiley-VCH GmbH

Various physical or chemical approaches have been explored to suppress lithium dendrites via physical blocking or regulating the nucleation/deposition process of lithium metal. Among these, electrically conductive hosts with high specific surface area were attached to Li metal, like reduced graphene oxide,[3] doped graphene,[4] and 3D porous skeletons,[5] to regulate lithium nucleation for a uniform electrode surface during lithium electrodeposition. There are also plenty of strategies of controlling dendrite structures, pathways and directions, including electrolyte additives,[6] artificial SEI protective layers,[7] high-mechanical-modulus solid-state electrolytes,[8] polymer electrolyte with high cationic transference number,[9,10] etc. All these approaches were dedicated to controlling the lithium deposition behavior and electrochemical reaction during battery operation, thereby obtaining optimized cycling CEs and reducing the risk of battery failure.

Apart from the abovementioned strategies, the formation of dendritic lithium has been revisited and associated with critical thermodynamic factors,[11–17] demonstrating that the thermodynamics parameters are important determinant for electrochemical reaction of lithium during battery operation.[12] Moreover, the fast-charging process of LMBs even brings acceleration of local heat generation and the rapid formation of local temperature "hotspots". This internal local high temperature can incense the local reaction rate, leading to preferable lithium nucleation and deposition on the hotspots to form lithium dendrites.[15,18] The roles of thermal field on regulating lithium structure, morphology, growth rate, and SEI structure have been studied via theoretical simulations.[19–22] The theoretical results predict that the uniform thermal field can effectively reduce the porosity of the deposited lithium and flatten the lithium morphology.[18] There are also some thermodynamic parameters other than thermal field to regulate lithium deposition.[23–25] However, these works of thermal management basically focus on regulating the thermal field as a whole in cells. Thus, there is minor attention paid to the local "hotspots" generated during irregular and dendritic lithium growth, to explore the relations between these "hotspots" and Li dendrite evolution.

In this work, a model of electrochemical lithium deposition concerted with a heat transfer model was established to probe the time-dependent power evolution of heat generation given various deposition overpotentials, showing the thermal and Li ion distribution in the lithium deposition system. As a result, the local hotspots were observed on the tips of protuberance area of irregular Li growth and incipient dendrites. The evident high temperature and Li ion flux gradient around the tip area, compared to that on the flat area of Li, aggravates the nonuniform irregular Li deposition and further Li dendrite growth. Considering the extremely high thermal conductivity of graphene flakes in the range of 3080–5150 W m^{-1} K^{-1} at room temperature,[26] a separator coated by a layer-stacked graphene film was constructed. The graphene side faces to Li metal, which functions as an in situ thermal dispersion medium to preclude local thermal accumulation around the irregular Li deposition and inhibit further dendrite growth. The Li‖Cu cell using graphene/PP composite separator shows dense and bulky Li deposition with high CE of >95% after 250 cycles at 1 mA cm^{-2}. Besides, the as-constructed separator recovers the Li metal from suffering a low CE of ≈65% to a high CE of >95% with prolonged cycle life. Li‖$LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ full cells with a low negative-to-positive (N/P) capacity ratio of ≈3.3 and low electrolyte-to-capacity (E/C) ratio of 4.4 µL mAh^{-1} using this separator deliver a high initial discharge specific capacity of 193 mAh g^{-1} and good capacity retention of 94% after 40 cycles at 0.2 C compared to the 48.3% capacity retention of cell with PP separator. Notably, the capacity of degraded cell with PP separator was recovered from 98.3 to 184.6 mAh g^{-1} with further cycling after reassembling with the composite separator. This works highlights that the thermodynamic factor of local hotspots in dendritic Li tip area is crucial in Li deterioration. The internal elimination of the local heat can be achieved by the as-constructed thermal conductive graphene/PP composite separator, which paves way for the development of high performance LMBs.

2. Results and Discussion

2.1. Thermodynamic Simulation of Li Deposition

The electrochemical deposition of lithium metal was simulated by coupling the electrochemical deposition module, involving tertiary current distribution and Nernst–Planck interface, with the heat transfer modules in COMSOL Multiphysics.[27,28] The geometric size and solution domain settings are illustrated in Figure S1 (Supporting Information), where the electrolyte domain adopts the thermodynamics parameters of traditional organic liquid electrolyte,[16] and the detailed simulation parameter settings are listed in Table S1 (Supporting Information). In this simulation, we present an initial protuberance on the center of Li metal electrode surface at the plating side representing the metallurgical defect of metallic foils. As the simulation progresses, Li ions diffuse and migrate under the collective driving force of electric field and ionic concentration gradient, then Li deposits to the electrode surface upon reduction.[29] The protuberance induces non-uniform distribution of electric field and large potential difference due to the "tip effect", which concentrates Li ions flux and accompanies large reaction current density at the local sites (Figure S2, Supporting Information).[30] As a result, the lithium nucleation and deposition rate at the bump sites is apparently higher than that at any other smooth site, which leads to increasing irregular Li and its further dendritic growth. Specifically, the maximum heat generation rate occurs at the tips of the Li dendrites along with their growth as shown in **Figure 1**. The dendrite length and heat generation rate slightly vary over time at the deposition overpotential of 0.3 V and the exchange current density of 5000 A m^{-2} (Figure 1a–c), as the maximum heat generation rate evolves from ≈6.5×10^6 W m^{-3} (500 s) to 8.4×10^6 W m^{-3} (1000 s) and 8.9×10^6 W m^{-3} (1500 s). However, when the deposition overpotential and exchange current density increase from 0.25 to 0.35 V (Figure 1f–i) and 1000 to 9000 A m^{-2} (Figure S3, Supporting Information), respectively, the maximum heat generation rate at dendrite tips remarkably rises, leading to aggravated dendrite growth. Although there is no evident change in Li dendrite length, there is increasing dendritic branches as function of deposition time and overpotential. They are prone to detach

© 2022 Wiley-VCH GmbH

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

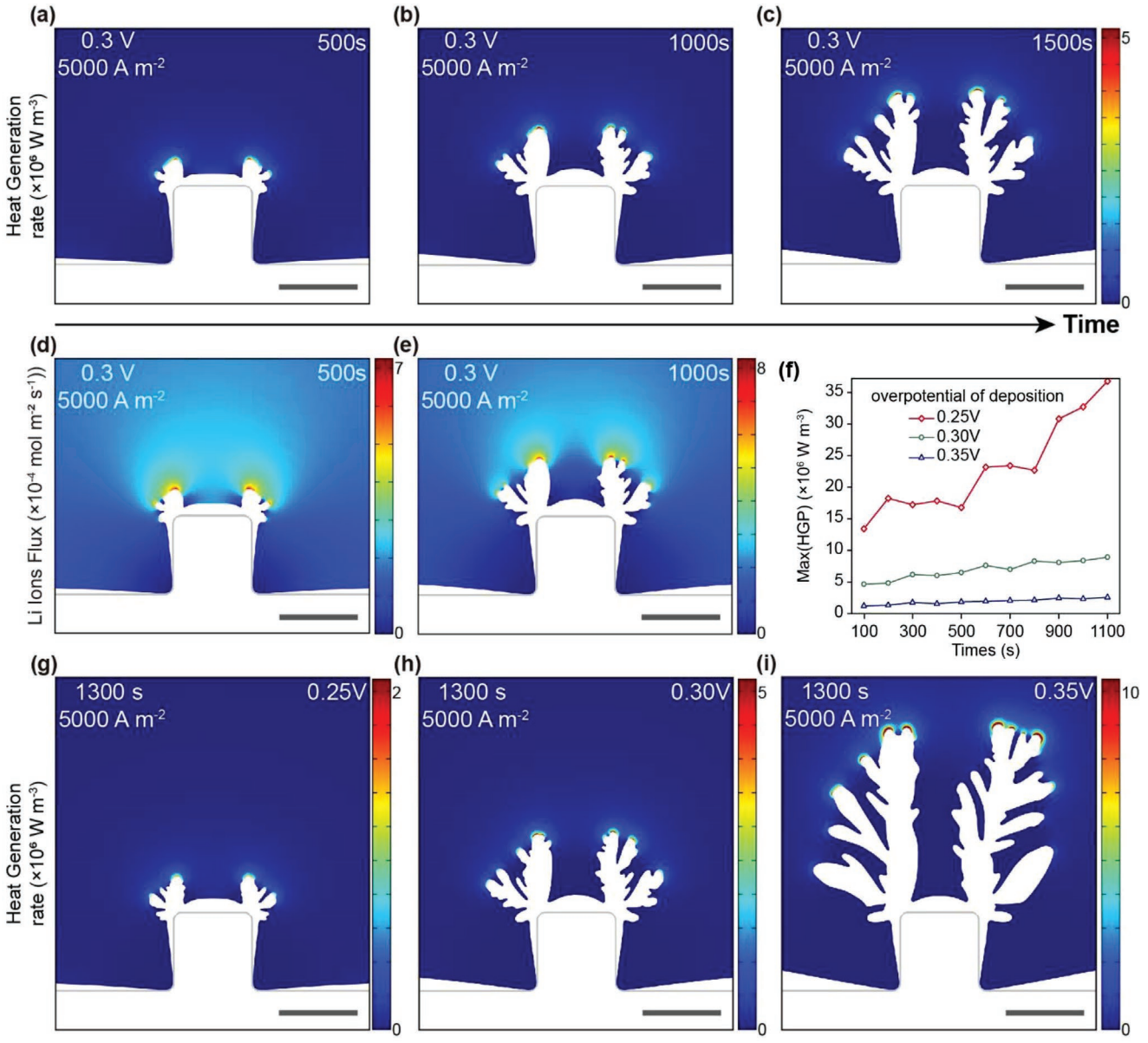


Figure 1. Li dendrite morphology evolution under thermodynamic deposition concerted with heat generation rate and lithium ions distribution in COMSOL. Dendrite growth evolution and heat generation rate around the protuberance at the time of a) 500 s, b) 1000 s, and c) 1500 s under the same deposition overpotential of 0.3 V, and under the deposition overpotential of g) 0.25 V, h) 0.3 V, and i) 0.35 V with the same evolution time of 1300 s and exchange current density of 5000 A m^{-2}. d,e) The corresponding Li ions flux distribution around the dendrite under 0.3 V in 500 and 1000 s, respectively. f) Plots of the maximum heat generation rate as a function of evolution time and the deposition overpotential. The gray lines in the simulation images represent the initial protuberance geometry before deposition. The scale bars are 10 µm.

from the Li metal substrate forming "dead lithium". More importantly, the overall temperature of the system gradually rises because of the ohmic heat and overpotential heat. The isotherm lines illustrate a relatively high temperature area ("hotspot") on the tips of dendritic Li (Figure S4, Supporting Information), which can be attributed to the locally high heat generation rate and poor thermal conductivity of conventional organic liquid electrolyte and polymer separator. These "local hotspots" can exacerbate further dendrite growth and have a self-amplified effect on local temperature rise and accelerated lithium deposition, owing to the concentrated lithium ion flux (Figure 1d,e) and enhanced exchange current density (Figure S5, Supporting Information).[15,31] Notably, the relatively loose dendrite structure also can generate great current heating effect,[32] which brings more severe dendrite issues and local temperature rise than that shown in our simulation. More seriously, the elevated temperature may cause contraction of separator, undesirable side-reactions of electrolyte, irreversible capacity fading, internal short circuit, and battery thermal runaway.[33]

© 2022 Wiley-VCH GmbH

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2.2. Materials Characterization of Thermally Conductive Composite Separator

To eliminate the dendrite-induced hotspots during lithium deposition, layer-stacked graphene flakes are coated on the polypropylene (PP) separators, acting as in situ thermal diffusion material during the electrochemical lithium deposition. This is owing to superior thermal conductivity especially the in-plane heat dispersion performance of graphene layer. The transmission electron microscopy (TEM) images (**Figure 2**a) show that exfoliated graphene sheets possess few-layer property. The graphene coating enhances the wetting and preserving ability of electrolyte, evident from the contact angle of 19.6° at the electrolyte/composite separator that is smaller than that of 40.9° for PP separator (Figure 2b,c). And the composite separator possesses comparable Li ion conductivity and transference number to that of the original separator (Figure S6, Supporting Information). As shown in the Raman spectrum of the composite separator (Figure 2e), two characteristic peaks of graphene are located at 1330 and 1580 cm^{-1}, corresponding the D band and G band, respectively.[34] The C1s X-ray photoelectron spectroscopy (XPS) (Figure S7, Supporting Information) of the as-prepared graphene layer exhibits the prominent C—C peaks at 284.2 eV, and the presence of C—OH (285.3 eV), C—O (286.5 eV), and C=O (288.1 eV) peaks, corresponding to the chemical composition and bonding configurations of electrochemically exfoliated graphene.[35,36] The surface of commercial PP separator with typical skeleton-pore structure (Figure S8, Supporting Information) is uniformly coated by 2D graphene sheets (Figure 2d) and the mechanical strength of the composite separator has been preserved very well after graphene coating (Figure S9, Supporting Information). This stable mechanical structure provides sufficient heat diffusion paths.[37] Further, a temperature distribution experiment (Figure 2g) was conducted to verify the improved thermal diffusion performance of composite separator (graphene@PP), which involved with a laser beam as heat input (0.5 W, 20 s) to create a hotspot on the surface of samples, and an infrared radiation (IR) camera to track the temperature

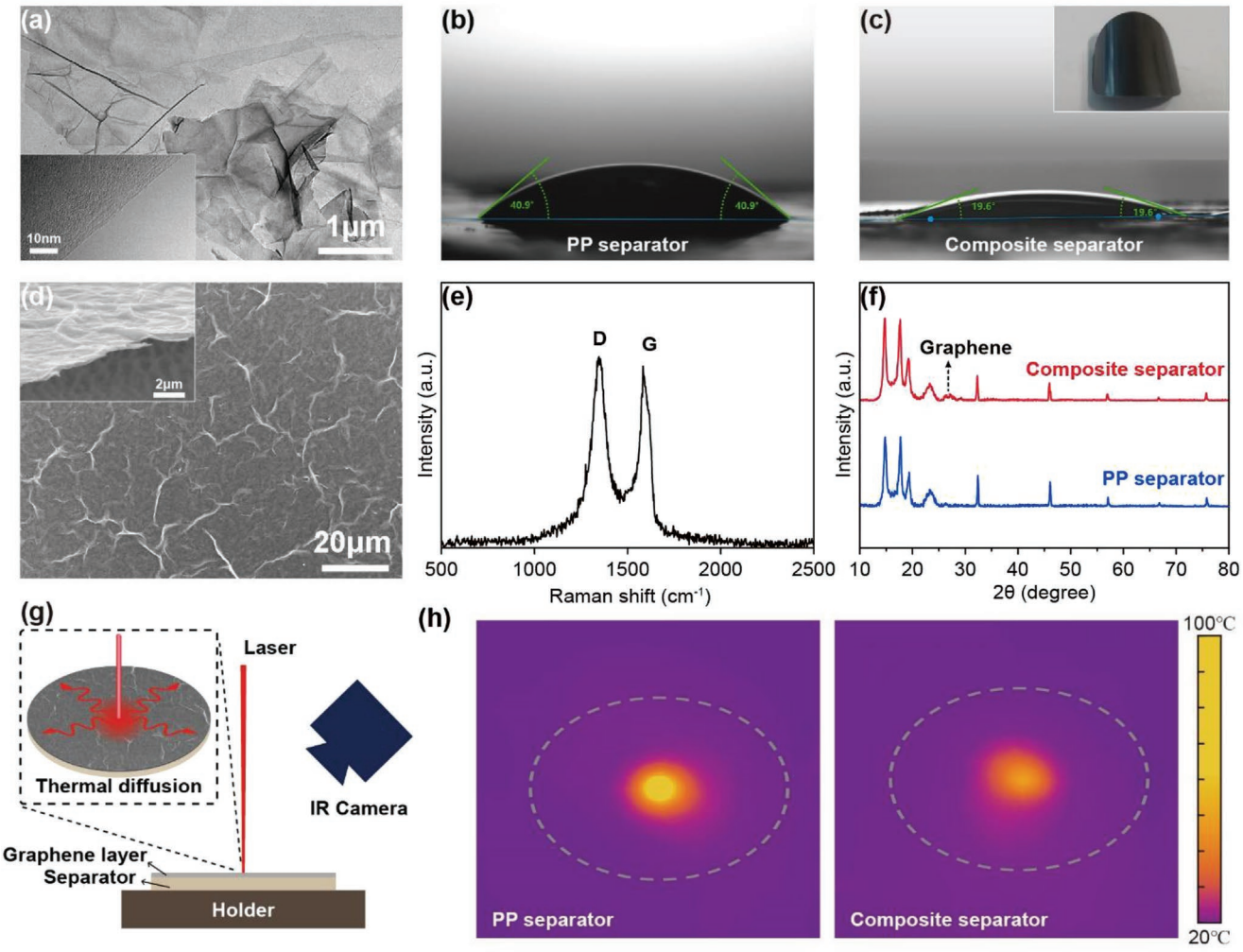


Figure 2. Characterizations of as-prepared graphene and composite separator. a) TEM images and magnified view of graphene. b,c) Contact angles of electrolyte on bare separator and composite separator, respectively. Inset: photo of the composite separator with graphene layer. d) SEM images of the graphene-coated separator in top-view and cross-section view (inset). e) Raman spectrum and f) XRD patterns of composite separator (graphene@PP) and PP separator. g) Schematic illustration of verifying the in-plane thermal diffusion performance of graphene layer with a focused laser as the hotspot input and the IR camera. h) The corresponding temperature distribution detection with pristine separator and composite separator.

© 2022 Wiley-VCH GmbH

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

evolution relating to the thermal dissipation along the plane. As shown in Figure 2h, the maximum temperature of the bare PP separator is about 78 °C, while a relatively low maximum temperature of about 53 °C was detected on the composite separator. It also presents much larger thermally affected area, which should be ascribed from the timely thermal diffusion by the layer-stacked 2D graphene sheets.[37,38] Thus, the composite separator is able to effectively eliminate the hotspots, whose application in lithium deposition process is foreseen to promote uniform lithium nucleation and growth based on its homogeneous thermal distribution of electrode surface (Figure 2h).

2.3. Lithium Deposition Behavior and Electrochemical Performance

To investigate the electrochemical deposition behavior of lithium, Li||Cu half cells were assembled by using lithium metal, Cu foil and commercial PP or composite separators as the counter/reference electrode, working electrode, and separator, respectively. Conventionally, extensive dendrite-like irregular lithium deposition can be observed on the Cu substrate after cycling using PP separator (**Figure 3**b,c).[39] In comparison, relatively uniform lithium deposition can be achieved in

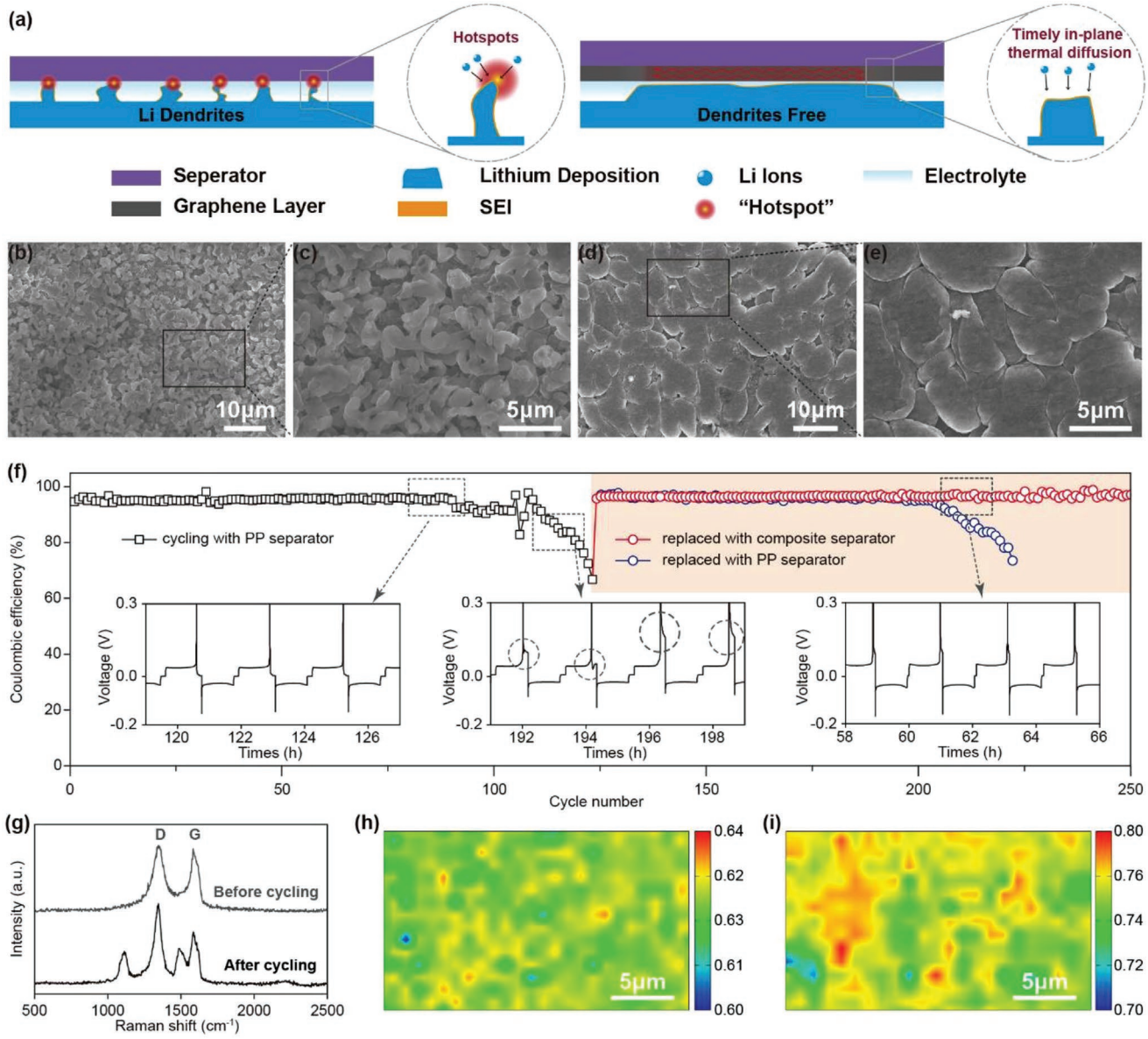


Figure 3. Study of lithium deposition behavior. a) Schematic of Li ion deposition under localized-temperature hotspots (left) and eliminated hotspots with reinforced thermal diffusion (right). SEM images of Li electrodeposition on Cu substrate under b) pristine PP separator and d) composite separator at a current density of 1 mA cm^{-2} with a capacity of 1 mAh cm^{-2} and c,e) the corresponding local magnification views, respectively. f) Cycling CEs of Li||Cu cells at 1 mA cm^{-2} and 1 mAh cm^{-2} with PP separator, reassembled with composite separator, or PP separator. The inset images were the time–voltage profiles at designated cycles. g) Raman spectra of composite separator before and after cycling. I_D/I_G intensity ratio mapping of composite separator h) before cycling and i) after cycling, respectively.

© 2022 Wiley-VCH GmbH

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

cells with composite separator benefitting from the enhanced thermal conductivity at the Li/graphene interface (Figure 3d,e). Such dense deposition can decrease the contact area between Li electrode surface and electrolyte (low Li porosity), which reduces the occurrence of side reaction of Li and electrolyte.[40] As illustrated in Figure 3a, because of the uneven ion distribution owing to the structural property of PP separator and original mechanical/chemical unevenness of the Cu substrate surface,[41,42] it tends to form lithium dendrite initially and keep growing by virtue of the localized high exchange current density and accumulated irreversible heat in the cells.[43] Besides, the "hotspots" can accelerate the local lithium ion enrichment around the formed Li tips, further aggravating the growth of lithium dendrites.[18] However, the graphene layer enables timely in-plane thermal diffusion presenting the effectiveness of suppressing dendrites deterioration (Figure 3a). The high and stable cycling CEs and charge–discharge curves of Li‖Cu cell with composite separator validate significant improvement compared to the unstable cycling of cells using PP separator (Figure S10, Supporting Information, and Figure 3f). Furthermore, the Li‖Cu cell in ether-based electrolyte using bare PP separator was dissembled after drastic CE decay to ≈60% with plentiful lithium dendrite (Figure S12a, Supporting Information). As shown in Figure 3f and Figure S11 (Supporting Information), the cell was reassembled with composite separator and recovered to ideal charge/discharge behavior with high cycling CEs of >95%. Besides, the cell reassembled with composite separator delivers lower plating/stripping overpotential and longer cycle life compared to that with PP separator (Figure S12b,c, Supporting Information). The obvious localized domains of bulky Li with larger deposition size are observed on top of the Li dendrites, owing to the elimination of local hotspots by in-plane thermal diffusion of graphene (Figure S13, Supporting Information), suggesting that the composite separator inhibits the preferential deposition on the tip area of Li dendrite. Such "recover effect" can also be observed in half cells cycling with carbonate-based electrolyte (Figure S14, Supporting Information). The deteriorating cycling stability with low CEs due to lithium dendrites is recovered to a much more stable cycling state with high CEs, since the heterogeneous lithium deposition behavior turns into uniform one with smoother Li electrode surface by eliminating the locally temperature hotspots. This demonstrates that the introduction of the thermally conductive graphene layer prevents the formed dendrites from deterioration and further electrode degradation, by eliminating the dendrite-induced temperature hotspots and homogenizing the Li ion deposition behavior. A bare PP separator was used as control to reassemble the cycled Li‖Cu cell, the inferior cycling performance verifies that the improvement of electrochemical performance is indeed originated from the introduction of thermal conductive layer rather than the replenishment of electrolyte. The Raman spectra (Figure 3g) and mapping (Figure 3h,i) of composite separator at pristine state and after cycling (washed with DOL) show the intensified D band and increased I_D/I_G intensity ratio, which reflects an increased degree of graphene defects indicating the localized hotspots suppressed by the graphene layer. The distributive evolution of graphene defects shows that more intensive lithium deposition reaction occurs at some localized interfacial sites of graphene layer. This is correlated to the local temperature hotspots generated by the localized lithium dendrite growth, because the lithium deposition rate is exponentially related to the thermal effect according to the Arrhenius equation. [12] Such result can also be observed in separator cycling with carbonate-based electrolyte, which reflect the ubiquitous existence of localized enhanced reaction sites of Li stripping/plating and local temperature hotspots in different electrolyte systems. (Figure S15, Supporting Information) The cycled composite separator (scanning electron microscopy [SEM] in Figure S16, Supporting Information) shows two additional peaks, related to the SEI compounds like Li_2O, etc.[44] This further illustrates that the interfacial graphene layer plays a pivotal role of achieving uniform Li deposition to Cu substrate by eliminating the temperature difference caused by local hotspots.

The Li‖Cu cells with composite separator present superior electrochemical performance with high CEs and long lifespan under different current densities of 1, 2, and 3 mA cm^{-2} with the areal capacity of 1 mAh cm^{-2} with 1 M LiTFSI in DOL/DME (1:1 vol%) with 2% $LiNO_3$ (**Figure 4**a) compared to those using PP separator. At a current density of 1 mA cm^{-2}, the same initial CEs of above 97% were observed for both cells, which is consistent with most reports.[2,31,40,41] However, the CE of cells using PP separator sharply drops to ≈70% after 120 cycles, and the fluctuation of CE values reflects the occurrence of "soft" short-circuits owing to the fragile lithium dendrite.[45] By contrast, the CEs of cells using composite separator remain at a high level of ≈95% even for 250 cycles, revealing excellent reversibility of lithium plating/stripping process. Similar tendency is also observed when the current density is increased to 2 and 3 mA cm^{-2} with the same deposition areal capacity of 1 mAh cm^{-2} (Figure 4a). The CEs of cells with PP separator decrease to 76% after 80 cycles under 2 mA cm^{-2}, and 62% after only 60 cycles under 3 mA cm^{-2}, respectively. Comparatively, the CEs of cells using composite separator maintain >95% under the same cycling condition. The effectiveness of improved thermal conductivity on alleviating dendrite deterioration still exists even under harsh condition of a current density of 5 mA cm^{-2} with 5 mAh cm^{-2} (Figure S17, Supporting Information) and an elevated temperature of 60 °C (Figure S18, Supporting Information). As shown in Figure S19 (Supporting Information), such electrochemical performance improvement is also observed in cells tested in carbonate-based electrolyte. These cells using composite separator show nearly constant CEs of 90% within 100 cycles at 1 mA cm^{-2} and 1 mAh cm^{-2}, despite the as-formed SEI is inferior to that formed under ether-based electrolyte causing low CEs and shorter cycles.[46]

The overpotential during Li plating and stripping at 1, 2 and 3 mA cm^{-2} is shown in Figure 4b, the voltage hysteresis is 24.4, 29.7, and 36.8 mV, respectively, for the cell with composite separator, compared to 25.1, 35.3, and 40.9 mV, respectively, for the bare PP separator. The similar minor electrode polarization illustrates that the Li ion diffusion is not disturbed by introducing the graphene layer, corresponding to its excellent electrolyte wettability and preservability (Figure 2c). The interface properties during cycling were further explored by electrochemical impedance spectroscopy (EIS) (Figure 4c). The spectra can be decoupled to interfacial resistance and bulk resistance according to the equivalent circuit model,[47,48] The interfacial

© 2022 Wiley-VCH GmbH

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

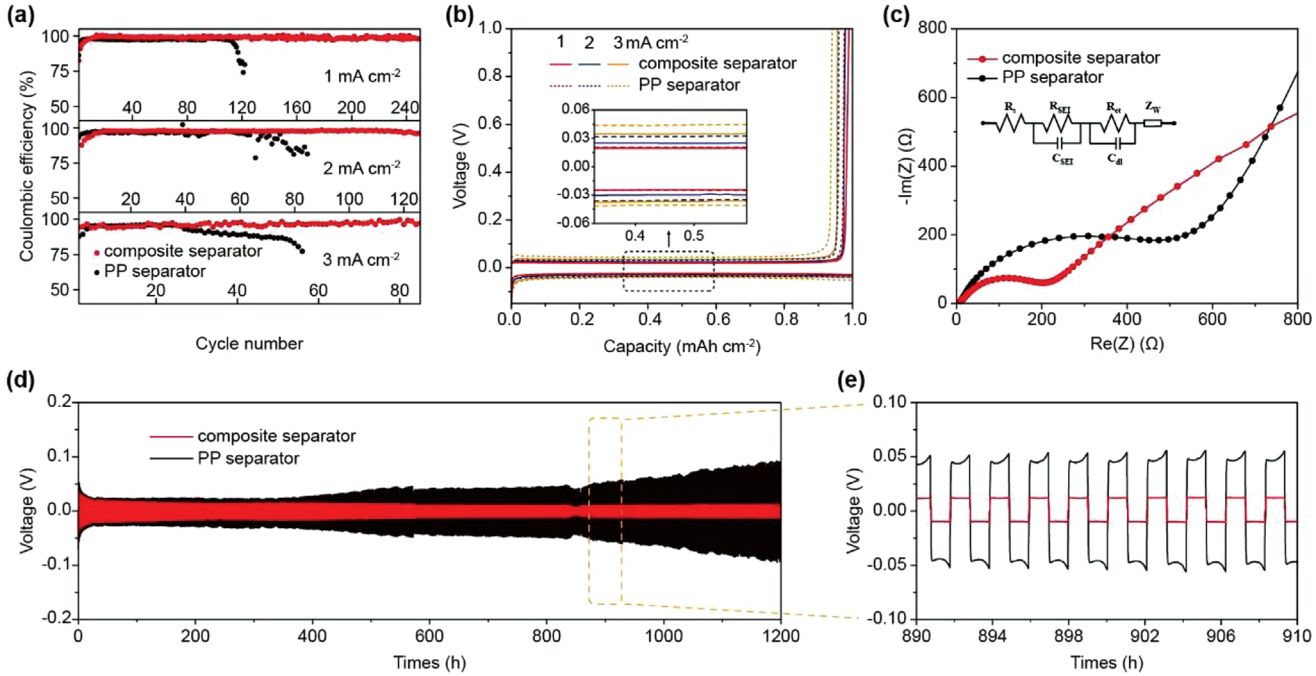


Figure 4. Electrochemical performance of cells with bare separator and composite separator. a) Cycling CEs of Li||Cu half cells with bare and composite separator at current densities of 1, 2, and 3 mA cm^{-2} with the same deposition capacity of 1 mAh cm^{-2} [electrolyte: 1 M LiTFSI in DOL/DME (1:1 vol%) with 2% $LiNO_3$]. b) The corresponding voltage hysteresis curves of Li plating/stripping for Li||Cu half cells with bare separator and composite separator at different current densities at the 20th cycle, respectively. Inset: magnified view of the bottom plot. c) Impedance spectra analysis of half cells after 100 cycles at 1 mA cm^{-2} and 1 mAh cm^{-2} with ether-based electrolyte and equivalent circuit diagram. d) Time–voltage curves of Li plating/stripping in symmetrical Li|| bare separator ||Li cell and Li|| composite separator ||Li cell at 1 mA cm^{-2} and 1 mAh cm^{-2} with 1 M LiTFSI in DOL/DME (1:1 vol%) with 2% $LiNO_3$ and e) the enlarged view of time–voltage curves from 890 to 910 h.

resistance of cells using composite separator are much lower than that of cells using PP separator, showing a value around 137 ohm after 100 cycles compared to the value of 306 ohm for PP separator. This characteristic interfacial resistance rise is related to the unstable SEI accumulation owing to increasingly severe lithium dendrite growth.[41] The long-term cycling and fast charging performance is further validated by Li||Li symmetrical cell with stripping/plating capacity of 1 mAh cm^{-2} at 1 mA cm^{-2} (Figure 4d) and 5 mAh cm^{-2} at 20 mA cm^{-2} (Figure S20, Supporting Information). As shown in Figure 4d,e, the overpotential of symmetric Li||PP separator||Li metal cell gradually increases during cycling from 20 to 90 mV, implying a deteriorating electrode/electrolyte interface caused by dendrite growth. However, the symmetric cells using the composite separator present stable cycling with low Li stripping/plating overpotential after over 1200 h at 1 mA cm^{-2} (Figure 4d,e) and 100 h at the current density of up to 20 mA cm^{-2} without obvious deterioration (Figure S20, Supporting Information), which meets the standards for high performance LMBs.[49,50]

2.4. Full Cell Performance of Lithium Metal Batteries

To validate the applicability of the design of local-hotspot eliminating separator, the effectiveness in full cells was explored by using high areal capacity $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM811, ≈6 mAh cm^{-2}) as cathode and lean lithium (100 µm-thick) as anode (low N/P ratio of 3.3) with lean electrolyte (E/C ratio of 4.4 µL mAh^{-1}). Above all, excessive lithium anode (600 µm-thick) was used to assemble the full cells with both separators, which shows approaching rate performance of the NCM811 cathode (Figure S21, Supporting Information). The cells deliver a discharge capacity of ≈205 mAh g^{-1} at 0.1 C (≈0.6 mA cm^{-2}) and retain charge/discharge ability when the current density increases to 2 C (≈12 mA cm^{-2}). Next, we evaluated the rate capability of full cells under lean lithium condition (100 µm-thick) with N/P capacity ratio of ≈3.3 from 0.1 to 2 C (**Figure 5**a). The full cell use composite separator delivers discharge capacity of 199, 184, and 115 mAh g^{-1} at 0.2, 0.5, and 1.0 C, respectively, which is superior to that of 197.5, 177, and 102 mAh g^{-1} using conventional PP separator. As the local heat generation tends to be more severe along with the incensement of charging rates,[13,16] the improved rate capability using the composite separator suggests that the local thermal nonuniformity can be effectively eliminated by the thermal conductive graphene layer. The cycling stability of full cells at 0.2 C rate is further demonstrated in Figure 5b and Figure S22 (Supporting Information). The cell with composite separator displays a remarkable capacity retention of 94% after 40 cycles with an initial discharge specific capacity of 188 mAh g^{-1} at 0.2 C (Figure S22b, Supporting Information), while the capacity of cell with PP separator with lean electrolyte severely declines to 100 mAh g^{-1} (50% of the initial) within the same cycling period (Figure S22c, Supporting Information). This results from the surface degradation of lithium metal anode. It is worth pointing out that there exist other challenges for achieving long-term cycling

© 2022 Wiley-VCH GmbH

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

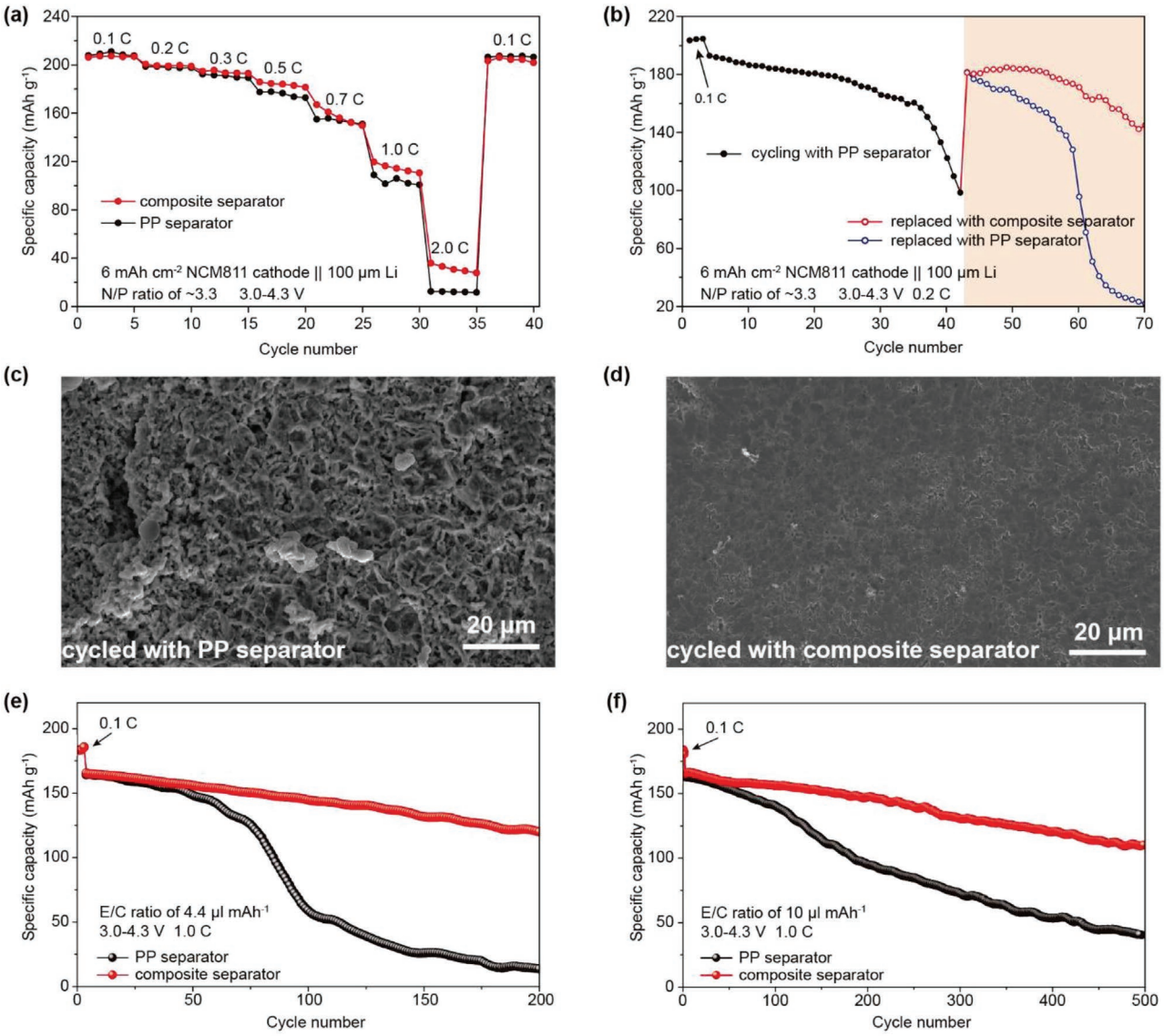


Figure 5. Electrochemical performance of Li||NMC811 cells with an N/P ratio of ≈3.3 in EC/DEC/DMC (1:1:1) electrolyte using 1 M $LiPF_6$ salt. a) Rate performance of full cells with different separators at various C-rates from 0.1 to 2 C. b) Cycling performance of the Li||NCM811 full cell cycling with PP separator and lean lithium (100 μm-thick) at 0.2 C, reassembled with composite separator or PP separator. The SEM images of cycled Li metal anode disassembled from Li||NCM811 full cells with c) pristine separator and d) composite separator. Cycling performance of the Li||NCM811 full cells cycling with PP separator or composite separator at 1.0 C with relative low material loading cathode of ≈2 mg cm^{-2} and e) low E/C ratio of 4.4 μL mAh^{-1} and f) E/C ratio of 10 μL mAh^{-1}.

LMBs with high Coulombic efficiency under low N/P ratio and lean electrolyte conditions (Table S2, Supporting Information). However, our work provides a unique thermodynamic perspective of dendrite-induced local hotspot to insight into the lithium dendrite evolution. Moreover, the capacity of cell with PP separator is recovered to a high level of 181 mAh g^{-1} after being reassembled with our composite separator and E/C ratio of 4.4 μL mAh^{-1}. It retains a high capacity of 171 mAh g^{-1} after another 30 cycles, which is superior to that of 20 mAh g^{-1} in cells reassembled with PP separator (Figure 5b). This full cell capacity recovery is identical with Li||Cu CE recovery in Figure 3f. The thermally conductive graphene layer takes effect on the surface of the lithium dendrites to ensure the uniformity of subsequent deposition by eliminating temperature hotspots. The post-mortem analysis of cycled Li metals was also performed shown in the SEM images (Figure 5c,d). A rough and porous surface with large amounts of needle-like dendrites can be observed on the surface of lithium electrode cycled with conventional PP separator owing to the nonuniform lithium deposition. By contrast, the lithium electrode cycled with composite separator displays a relatively smooth morphology, demonstrating the thermally conductive graphene layer eliminates the temperature hotspots and deter the further dendrites deterioration and even Li electrode degradation.

Limited N/P ratio and E/C ratio are critical for fair performance comparison and practical battery production, but they

© 2022 Wiley-VCH GmbH

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

may greatly limit the electrochemical performance especially the cycling life. Lean electrolyte is not only consumed by the inevitable SEI formation on the lithium metal anode but also decomposed at the sensitive surface of the Ni-rich NCM811 cathode particles (surface parasitic reactions).[51] To further investigate the critical role of N/P ratio and E/C ratio on the cycling life, the cells were assembled and evaluated using a normal cathode loading of 2 mg cm^{-2} (high N/P ratio of ≈50) with electrolyte usage (lower E/C ratio of 4.4 µL mAh^{-1} and higher one of 10 µL mAh^{-1}). As shown in Figure 5e,f, all the cells with the composite separator present superior cycling life to the cells with PP separator.

In general, there exist lithium dendrites on the surface of lithium metal anode during repeated stripping and plating process owing to the nonuniform Li electrochemical deposition. The locally temperature hotspots on the dendrite tips induces Li ions to deposit preferentially on the dendrite tips and exacerbates further dendrite growth. However, the local hotspots are eliminated by rapid in-plane thermal diffusion of graphene layer in the composite separator. Thus, much more uniform Li deposition can be achieved to afford a smoother surface morphology without further dendrite growth, thereby recovering the Li electrode performance.

3. Conclusions

In summary, a coupling model of local heat transfer and the lithium electrochemical deposition was established in COMSOL to investigate the evolution of heat generation rate around lithium dendrites and its correlation with local dendritic growth. It is unveiled that the overpotential heat generated by locally accelerated electrochemical deposition reaction is prone to accumulate at the deposition tip area to form local temperature "hotspots", which stems from the metallurgical defect and thermodynamically unstable Li-anode/electrolyte interface. This results in subsequent fast dendrite growth and, in turn, causes more serious hotspots extremes, due to the poor thermal conductivity of commercial cell components. Increasing the thermal conductivity on the Li metal surface to timely diffuse the local heat generated by dendrite growth is an effective way to alleviate the dendrite growth to large extent. Our design of graphene-layer-coated separator acts as the role of "heatsink" to eliminate the emergence of local "hotspots" in the initial stage of dendrite formation and preclude potential risk of fast dendrite deterioration. Cells with composite separator show superior rate capability and prolonged cycle life span with relatively smooth lithium deposition morphology and stable interface during cycling. Notably the composite separator can be used to recover the cells suffering from drastic decay. This work provides a unique thermodynamic perspective of dendrite-induced local hotspot to insight into the lithium dendrite evolution, paving way for effective Li metal protection to boost the practical application of LMBs.

4. Experimental Section

Simulation: The finite element method simulation of lithium electrodeposition considering thermodynamic characteristics was performed with COMSOL Multiphysics. The electrodeposition multiphysics module combining the tertiary current distribution, Nernst–Planck, and the deformed geometry was chosen to simulate the lithium electrochemical deposition process on the surface of Li metal anode. The heat transfer in solids and fluids module was selected to explore the thermal effect on lithium dendrite growth. In the electrodeposition module, the Li ion transport behavior in electrolyte including diffusion caused by concentration gradient and electromigration induced by electric filed was governed by the Nernst–Planck equation and Poisson equation, and the impact of convection is neglected. The mass transfer in electric field follows the Nernst–Einstein Relation to obtain the mobility relating to the diffusion coefficient in electrolyte. The concentration-dependent kinetics equation was adopted to describe the electrochemical reaction on the electrode surface. Besides, deformed geometry submodule updates the electrode surface morphology according to the normal velocity, which calculated from the lithium deposition current density and re-mashing the mesh when necessary.

The normal velocity (υ_n) of the surface is given by

$$\upsilon_n = -\frac{\Omega_{Li}}{F} i_{Li} \tag{1}$$

where Ω_{Li}, i_{Li}, and F are the Li atom molar volume, reaction current density, and Faraday constant, respectively. In heat transfer submodule, the temperature distribution is obtained by the energy equation, where Cp, ρ, and k are the specific heat capacity, mass density, and thermal conductivity of electrode and electrolyte, respectively. The heat generation rate Q only considers the reversible heat, including the Ohmic heat (Q_{ohmic}) owing to the ionic resistance, and overpotential heat (Q_{op}) relating to the overpotential and reaction current density at the electrode/electrolyte interface.

$$Q = Q_{ohmic} + Q_{op} \tag{2}$$

$$Q_{ohmic} = \sigma_s (\nabla \phi_s)^2 + \sigma_l (\nabla \phi_l)^2 \tag{3}$$

$$Q_{op} = a_s j \phi \tag{4}$$

where σ_s, σ_l, ϕ_s, ϕ_l, a_s, j, and ϕ are the conductivity in electrode, conductivity in electrolyte, electrostatic potential in electrode, electrostatic potential in electrolyte, specific surface area per unit volume of the electrode, current density, and overpotential in the electrode/electrolyte interface, respectively.

The initial temperature $T = T_0$ and convection and radiation boundary conditions were applied to the boundary of the system to solve the heat transfer field. Most importantly, a temperature-dependent ion diffusion coefficient (D_l) expression was introduced to the simulation to link the electrodeposition model and heat transfer model.

$$D_l = a_0 \exp\left[-bc + \frac{E_D}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right] \tag{5}$$

where a_0, b, C, E_D, R, and T are the temperature-independent pre-exponent, activation energy, molar gas constant, and temperature field, respectively.

Preparation of Materials: The graphene used in experiment was prepared by electrochemical exfoliation method. Potentiostatic DC electrolysis chose the graphite foil and platinum as electrode and $(NH_4)_2SO_4$ aqueous solution as electrolyte. The obtained mixture was suction filtered, washed, and re-dispersed in N-methyl-2-pyrrolidone (NMP), which was further centrifuged after standing for 24 h to obtain graphene dispersion in NMP solvent. The graphene in NMP suspension was coated on one side of the Celgard 2500 separator through a simple vacuum filtration with an area loading mass of 0.16 mg cm^{-2}. The composite separator was dried in a vacuum oven at 60 °C for 24 h and subsequently cut into 18 mm diameter wafers for coin cells assembly. The ternary cathodes were synthesized from 95 wt% $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$

 © 2022 Wiley-VCH GmbH

powders, 2.5 wt% super P carbon blacks, and 2.5 wt% PVDF binder. The three components were fully mixed in NMP by magnetic stirring and the slurry was coated on an aluminum foil using doctor blade methods. The electrodes were firstly blast dried for 2 h to fast drying and then vacuum dried for 48 h to remove solvent completely. The NMC811 cathode foil was punched into 14 mm circular disks for coin cells and the active material loading was about 30.06 mg cm^{-2}.

Electrochemical Tests: The CR2032-type Li‖Cu, Li‖Li, and Li‖NMC811 coin cells were assembled with Li metal foil (0.1 or 0.6 mm thick) as the reference electrode in an Ar-filled glovebox with O_2 and H_2O content <0.1 ppm. The carbonate-based electrolyte used in the cells was 1 M $LiPF_6$ in EC/DEC/DMC (1:1:1 vol%) and the ether-based electrolyte was 1 M LiTFSI in DOL/DME (1:1 vol%) with 2% $LiNO_3$ (by weight) additive. The amount of electrolyte injected into each cell was obtained by calculating the corresponding E/C ratio. The graphene layer was placed to face to the bare Cu electrode in the Li‖Cu cells and to the Li metal anode in the Li‖NMC811 cells.

All of the charge/discharge cycling tests of the cells were all performed by battery test equipment (Neware). In Li‖Cu cells, the area capacity of 1 mAh cm^{-2} Li was deposited on bare Cu electrode with different current density and set to be totally stripped in one cycle with the voltage of cell rise to 1 V in 1 M LiTFSI in DOL/DME (1:1 vol%) with 2% $LiNO_3$. The Li‖Li symmetric cells were set to charge at a current density of 1 mA cm^{-2} for 1 h and discharge under the same condition for one cycle. The Li‖NMC811 cells were cycled with a voltage range of 3.0–4.3 V. The resistance analysis was tested by electrochemical impedance spectroscopy measurement (a frequency range of 1 mHz to 100 kHz with an amplitude of 10 mV) with a Princeton electrochemical workstation.

Characterizations: To study the surface morphology of the cycled Li metal anodes and Cu current collector, the coin cells were dissembled, and then the cycled electrode was rinsed in pure DME or DEC to remove the residual electrolyte and Li salt and dried overnight at room temperature. All processes were carried out in the glovebox to avoid electrode exposure to atmosphere. The morphology of the separator and electrode was characterized on a MAIA3 LMH SEM operated at 5.0 kV and a JEOL IEM-F200 HRTEM. Raman spectra were recorded with a Laser Raman Spectrometer (523 nm wavelength laser). The element and groups analysis were obtained using XPS (Thermo Fisher ESCALAB). The structural properties of the separators were characterized by X-ray diffraction (XRD, Bruker D8 ADVANCE) with Cu Kα radiation. The contact angles were measured using DSA100.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

D.H., X.W., and Y.Z. contributed equally to this work. The authors gratefully acknowledge the support from the National Key R&D Program of China (2021YFB2400400), the National Natural Science Foundation of China (Grant No. 21905220), the China Postdoctoral Science Foundation (2020M673408), the Fundamental Research Funds for "Young Talent Support Plan" of Xi'an Jiaotong University (HG6J003), the Fundamental Research Funds for the Central Universities (xzy022021008), and the "1000-Plan program" of Shaanxi Province and the Jiangsu Provincial Department of Science and Technology (BK20201190). The authors also thank Instrument Analysis Center of Xi'an Jiaotong University for necessary characterizations and analysis. This work was supported by Singapore A*STAR's Central Research Funds (CRF) Award (Project No. SC25/21-111312).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

electrochemical-heat coupled simulation, graphene, Li metal anodes, local hotspots, thermally conductive membranes

Received: April 7, 2022
Published online: May 20, 2022

[1] I. R. M. Kottegoda, Y. Kadoma, H. Ikuta, Y. Uchimoto, M. Wakihara, *Electrochem. Solid-State Lett.* **2002**, *5*, A275.
[2] X. Zhang, A. Wang, X. Liu, J. Luo, *Acc. Chem. Res.* **2019**, *52*, 3223.
[3] D. Lin, Y. Liu, Z. Liang, H. W. Lee, J. Sun, H. Wang, K. Yan, J. Xie, Y. Cui, *Nat. Nanotechnol.* **2016**, *11*, 626.
[4] R. Zhang, X. R. Chen, X. Chen, X. B. Cheng, X. Q. Zhang, C. Yan, Q. Zhang, *Angew. Chem., Int. Ed.* **2017**, *56*, 7764.
[5] H. Liu, E. Wang, Q. Zhang, Y. Ren, X. Guo, L. Wang, G. Li, H. Yu, *Energy Storage Mater.* **2019**, *17*, 253.
[6] V.-C. Ho, D. T. Ngo, H. T. T. Le, R. Verma, H.-S. Kim, C.-N. Park, C.-J. Park, *Electrochim. Acta* **2018**, *279*, 213.
[7] P. Li, X. Dong, C. Li, J. Liu, Y. Liu, W. Feng, C. Wang, Y. Wang, Y. Xia, *Angew. Chem., Int. Ed.* **2019**, *58*, 2093.
[8] X. Zhang, S. Wang, C. Xue, C. Xin, Y. Lin, Y. Shen, L. Li, C. W. Nan, *Adv. Mater.* **2019**, *31*, 1806082.
[9] Z. Li, W. Lu, N. Zhang, Q. Pan, Y. Chen, G. Xu, D. Zeng, Y. Zhang, W. Cai, M. Yang, Z. Yang, Y. Sun, H. Ke, H. Cheng, *J. Mater. Chem. A* **2018**, *6*, 14330.
[10] J. Zhu, Z. Zhang, S. Zhao, A. S. Westover, I. Belharouak, P. F. Cao, *Adv. Energy Mater.* **2021**, *11*, 2003836.
[11] H. H. Yan, Y. H. Bie, X. Y. Cui, G. P. Xiong, L. Chen, *Energy Convers. Manage.* **2018**, *161*, 193.
[12] X. Gao, Y.-N. Zhou, D. Han, J. Zhou, D. Zhou, W. Tang, J. B. Goodenough, *Joule* **2020**, *4*, 1864.
[13] K. R. Adair, M. N. Banis, Y. Zhao, T. Bond, R. Li, X. Sun, *Adv. Mater.* **2020**, *32*, 2002550.
[14] J. Wang, W. Huang, A. Pei, Y. Li, F. Shi, X. Yu, Y. Cui, *Nat. Energy* **2019**, *4*, 664.
[15] H. Wang, Y. Zhu, S. C. Kim, A. Pei, Y. Li, D. T. Boyle, H. Wang, Z. Zhang, Y. Ye, W. Huang, Y. Liu, J. Xu, J. Li, F. Liu, Y. Cui, *Proc. Natl. Acad. Sci. USA* **2020**, *117*, 29453.
[16] Z. Hong, V. Viswanathan, *ACS Energy Lett.* **2019**, *4*, 1012.
[17] R. Akolkar, *J. Power Sources* **2014**, *246*, 84.
[18] B. S. Vishnugopi, F. Hao, A. Verma, P. P. Mukherjee, *ACS Appl. Mater. Interfaces* **2020**, *12*, 23931.
[19] C. T. Love, O. A. Baturina, K. E. Swider-Lyons, *ECS Electrochem. Lett.* **2014**, *4*, A24.
[20] K. Yan, J. Wang, S. Zhao, D. Zhou, B. Sun, Y. Cui, G. Wang, *Angew. Chem., Int. Ed.* **2019**, *58*, 11364.
[21] G. Liu, W. Lu, *J. Electrochem. Soc.* **2017**, *164*, A1826.
[22] Z. Hong, V. Viswanathan, *ACS Energy Lett.* **2018**, *3*, 1737.
[23] X.-G. Yang, T. Liu, C.-Y. Wang, *Nat. Energy* **2021**, *6*, 176.
[24] Y. Liu, Y. Qiao, Y. Zhang, Z. Yang, T. Gao, D. Kirsch, B. Liu, J. Song, B. Yang, L. Hu, *Energy Storage Mater.* **2018**, *12*, 197.
[25] R. Carter, C. T. Love, *ACS Appl. Mater. Interfaces* **2018**, *10*, 26328.
[26] S. Ghosh, I. Calizo, D. Teweldebrhan, E. P. Pokatilov, D. L. Nika, A. A. Balandin, W. Bao, F. Miao, C. N. Lau, *Appl. Phys. Lett.* **2008**, *92*, 151911.
[27] X. Guo, Y. Ding, H. Gao, J. B. Goodenough, G. Yu, *Adv. Mater.* **2020**, *32*, 2000316.

© 2022 Wiley-VCH GmbH

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[28] G. Zheng, C. Wang, A. Pei, J. Lopez, F. Shi, Z. Chen, A. D. Sendek, H.-W. Lee, Z. Lu, H. Schneider, M. M. Safont-Sempere, S. Chu, Z. Bao, Y. Cui, *ACS Energy Lett.* **2016**, *1*, 1247.
[29] G. Yoon, S. Moon, G. Ceder, K. Kang, *Chem. Mater.* **2018**, *30*, 6769.
[30] J. Tan, A. M. Tartakovsky, K. Ferris, E. M. Ryan, *J. Electrochem. Soc.* **2015**, *163*, A318.
[31] X. B. Cheng, T. Z. Hou, R. Zhang, H. J. Peng, C. Z. Zhao, J. Q. Huang, Q. Zhang, *Adv. Mater. Interfaces* **2016**, *28*, 2888.
[32] L. Kong, Y. Xing, M. G. Pecht, *IEEE Access* **2018**, *6*, 8387.
[33] S. Zheng, L. Wang, X. Feng, X. He, *J. Power Sources* **2018**, *378*, 527.
[34] L. M. Malard, M. A. Pimenta, G. Dresselhaus, M. S. Dresselhaus, *Phys. Rep.* **2009**, *473*, 51.
[35] W. K. Shin, A. G. Kannan, D. W. Kim, *ACS Appl. Mater. Interfaces* **2015**, *7*, 23700.
[36] K. Parvez, Z. S. Wu, R. Li, X. Liu, R. Graf, X. Feng, K. Mullen, *J. Am. Chem. Soc.* **2014**, *136*, 6083.
[37] Z. Wei, Z. Ni, K. Bi, M. Chen, Y. Chen, *Carbon* **2011**, *49*, 2653.
[38] K. R. Pyun, S. H. Ko, *Mater. Today Energy* **2019**, *12*, 431.
[39] Y. Chen, X. Dou, K. Wang, Y. Han, *Adv. Energy Mater.* **2019**, *9*, 1900019.
[40] Z. Huang, G. Zhou, W. Lv, Y. Deng, Y. Zhang, C. Zhang, F. Kang, Q.-H. Yang, *Nano Energy* **2019**, *61*, 47.
[41] C. Z. Zhao, P. Y. Chen, R. Zhang, X. Chen, B. Q. Li, X. Q. Zhang, X. B. Cheng, Q. Zhang, *Sci. Adv.* **2018**, *4*, eaat3446.
[42] A. Jana, D. R. Ely, R. E. García, *J. Power Sources* **2015**, *275*, 912.
[43] Y. Zhu, J. Xie, A. Pei, B. Liu, Y. Wu, D. Lin, J. Li, H. Wang, H. Chen, J. Xu, A. Yang, C. L. Wu, H. Wang, W. Chen, Y. Cui, *Nat. Commun.* **2019**, *10*, 2067.
[44] J. Nanda, G. Yang, T. Hou, D. N. Voylov, X. Li, R. E. Ruther, M. Naguib, K. Persson, G. M. Veith, A. P. Sokolov, *Joule* **2019**, *3*, 2001.
[45] D. Kang, N. Hart, J. Koh, L. Ma, W. Liang, J. Xu, S. Sardar, J. P. Lemmon, *Energy Storage Mater.* **2020**, *24*, 618.
[46] C. Fang, J. Li, M. Zhang, Y. Zhang, F. Yang, J. Z. Lee, M. H. Lee, J. Alvarado, M. A. Schroeder, Y. Yang, B. Lu, N. Williams, M. Ceja, L. Yang, M. Cai, J. Gu, K. Xu, X. Wang, Y. S. Meng, *Nature* **2019**, *572*, 511.
[47] J. Cui, S. Yao, M. Ihsan-Ul-Haq, J. Wu, J.-K. Kim, *Adv. Energy Mater.* **2019**, *9*, 1802777.
[48] J. R. Rodriguez, P. J. Kim, K. Kim, Z. Qi, H. Wang, V. G. Pol, *J. Colloid Interface Sci.* **2021**, *583*, 362.
[49] W. Chen, Y. Hu, W. Lv, T. Lei, X. Wang, Z. Li, M. Zhang, J. Huang, X. Du, Y. Yan, W. He, C. Liu, M. Liao, W. Zhang, J. Xiong, C. Yan, *Nat. Commun.* **2019**, *10*, 4973.
[50] Y. Fang, Y. Zhang, K. Zhu, R. Lian, Y. Gao, J. Yin, K. Ye, K. Cheng, J. Yan, G. Wang, Y. Wei, D. Cao, *ACS Nano* **2019**, *13*, 14319.
[51] H.-H. Ryu, K.-J. Park, C. S. Yoon, Y.-K. Sun, *Chem. Mater.* **2018**, *30*, 1155.

© 2022 Wiley-VCH GmbH

16146840, 2022, 25, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/aenm.202201190 by Sichuan University, Wiley Online Library on [06/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License